

# The Linear Model

## OLS, Inference, and Diagnostics

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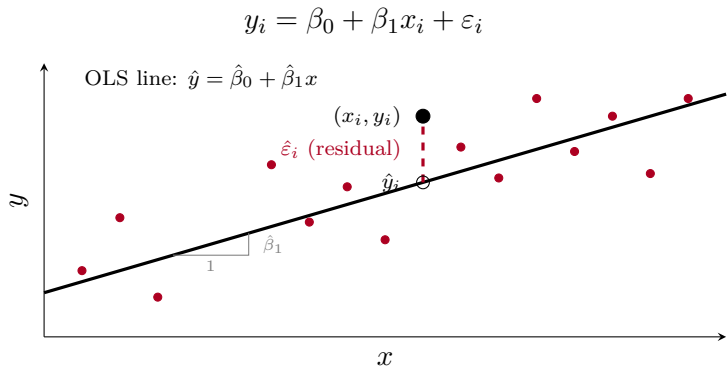
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# Where OLS comes from: Gauss and Ceres

- **1801** — Giuseppe Piazzi spots **Ceres** (a dwarf planet in the asteroid belt), then loses it behind the Sun after only a handful of noisy observations.
- **The question:** *where will it reappear?*
- The young **Carl Friedrich Gauss** computes the orbit with a new method — choose the trajectory that **minimises the sum of squared residuals** (the gaps between the observed positions and the predicted ones).
- **Result:** an accurate prediction; Ceres is found again. **Ordinary Least Squares (OLS)** is born.

# OLS in one picture



OLS picks  $(\hat{\beta}_0, \hat{\beta}_1)$  by **least squares** — minimising:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \underbrace{\sum_i \underbrace{(y_i - \beta_0 - \beta_1 x_i)}_{\hat{\varepsilon}_i}}_{\text{sum of squared residuals}}^2.$$

# Solving for $\hat{\beta}$ (simple regression)

Minimising the sum of squared residuals

$S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$ , set its two derivatives to zero:

$$\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \quad \sum_i x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

- The first gives  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$  — the line passes through  $(\bar{x}, \bar{y})$ .
- Substituting into the second and solving for the slope:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}.$$

( $\bar{x} = \frac{1}{n} \sum_i x_i$  is the sample mean; a hat marks an estimate. The  $1/n$  in Cov and Var cancels, so this ratio of moments equals the ratio of sums above.)

# Multiple regression

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \varepsilon_i$$

- Here  $k$  is the **number of explanatory variables** (slopes); with the intercept there are  $k + 1$  coefficients.
- Each  $\beta_j$  is the effect of  $x_j$  **holding the other regressors fixed** — identified from the variation in  $x_j$  *conditional on* the others.
- OLS minimises  $\sum_i (y_i - x'_i \beta)^2$ , with closed-form solution

$$\hat{\beta} = (X'X)^{-1}X'y$$

where  $X$  is the  $n \times (k+1)$  matrix of regressors and a **prime** denotes the **transpose** (so  $X'$  is  $X$  flipped — rows become columns).

# OLS by hand: a worked example

A small case — **3 observations, 2 parameters** — you can check on the board. Model  $y = \beta_0 + \beta_1 x$  with points  $(x, y) = (1, 2), (2, 2), (3, 5)$ :

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad y = \begin{pmatrix} 2 \\ 2 \\ 5 \end{pmatrix}, \quad X'X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \quad X'y = \begin{pmatrix} 9 \\ 21 \end{pmatrix}.$$

$$(X'X)^{-1} = \frac{1}{6} \begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}, \quad \hat{\beta} = (X'X)^{-1}X'y = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1.5 \end{pmatrix}.$$

- First column of  $X$  is **1s** (the intercept);  $\det(X'X) = 42 - 36 = 6$ .
- **Check:**  $\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{3}{2} = 1.5$  — the matrix formula reproduces the simple-regression slope. (More regressors  $\Rightarrow X$  gains columns; need  $n \geq k + 1$ .)

We can already *compute*  $\hat{\beta}$ . The rest of the lecture asks four questions:

- ➊ **Interpret** — what does a coefficient *mean*?
- ➋ **Inference** — is it *real* (precise, significant)?
- ➌ **Assumptions** — **and when they fail** — is it *valid* (unbiased, honest SE)?
- ➍ **Doing it in Python** (pandas + statsmodels).

## Section 1

### Interpreting coefficients



# Interpretation of coefficients

| model       | equation                          | a change in $x$ ... | ... changes $y$ by        |
|-------------|-----------------------------------|---------------------|---------------------------|
| level-level | $y = \beta_0 + \beta_1 x$         | +1 unit             | $\beta_1$ units           |
| log-level   | $\ln y = \beta_0 + \beta_1 x$     | +1 unit             | $100(e^{\beta_1} - 1) \%$ |
| level-log   | $y = \beta_0 + \beta_1 \ln x$     | +1%                 | $\beta_1/100$ units       |
| log-log     | $\ln y = \beta_0 + \beta_1 \ln x$ | +1%                 | $\beta_1 \%$ (elasticity) |

For small  $\beta_1$  these collapse to the familiar approximations (e.g. log-level  $\approx 100\beta_1\%$ ); the exact forms matter only when  $\beta_1$  is sizeable.

## Example: the gender wage gap

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{woman}_i + \varepsilon_i, \quad \hat{\beta}_1 \approx -0.285.$$

Women earn on average  $\approx$  **24.8%** less than men (INSEE, 2017). For a log outcome with a **dummy** regressor, use the exact effect:

$$e^{\hat{\beta}_1} - 1 = e^{-0.285} - 1 = -0.248 = -24.8\%.$$

# Ceteris paribus

Add controls — education, experience, hours, occupation, sector:

$$\ln(\text{wage}_i) = \beta_0 + \beta_1 \text{woman}_i + \beta_2 \text{educ}_i + \cdots + \varepsilon_i, \quad \hat{\beta}_1 \approx -0.10.$$

- “**All else equal**,” women earn  $\approx 9.5\%$  less — exactly  $e^{-0.10} - 1 = -0.095$ . The gap shrinks once we hold **observables** fixed, but only the *observed* ones.
- **Careful — not every control helps**. A variable that is itself a *consequence* of being a woman (occupation, sector — if discrimination channels women into lower-paid roles) is a **bad control**: adding it can *hide* part of the true gap. Which controls to include is already a causal question.

## What's left to check

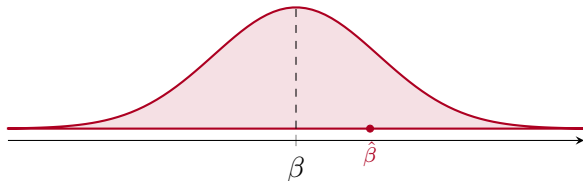
We have read these coefficients at face value. Two questions remain: are they statistically distinguishable from zero (**inference**, next), and are they unbiased in the first place (**assumptions**, after)?

## Section 2

### Inference

# The true $\beta$ we never observe

- Behind the data is a **population** (a data-generating process) with a **true but unknown** coefficient  $\beta$  — the effect of  $x$  on  $y$  for *everyone*. We never see it.
- From **one sample** of  $n$  units we compute an **estimate**  $\hat{\beta}$ . A *different* sample would give a *different*  $\hat{\beta}$  — so  $\hat{\beta}$  is **random**, with a **sampling distribution**:



One sample gives one draw  $\hat{\beta}$  from this curve. **Inference** asks how far it is likely to land from the true  $\beta$ . (*The curve is drawn centred at  $\beta$  — that OLS is unbiased is the next slide.*)

## How precise is $\hat{\beta}$ ? The standard error

- OLS is **unbiased** (and **consistent**): across samples  $\hat{\beta}$  is centred on the truth,  $\mathbb{E}(\hat{\beta}) = \beta$ .
- The **standard error** is the **standard-deviation spread** of that curve,  $\widehat{\text{se}}(\hat{\beta}) = \sqrt{\widehat{\text{Var}}(\hat{\beta})}$  — how far a typical  $\hat{\beta}$  lands from  $\beta$ .
- In simple regression  $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$ , with  $\sigma^2 = \text{Var}(\varepsilon_i)$  the error variance (estimated by  $\hat{\sigma}^2 = \text{RSS}/(n - k - 1)$ , with RSS the **residual sum of squares**  $\sum_i \hat{\varepsilon}_i^2$ ). So the SE is **smaller** when  $n$  is large, the regressor varies more ( $\sum_i (x_i - \bar{x})^2 = n \text{Var}(x)$ ), or residuals are tight.

$n - k - 1$  is the **degrees of freedom** — we “use up”  $k+1$  observations estimating the coefficients. This is the *classical* SE (valid under homoskedasticity); we will see it can be wrong, and how to fix it (robust SE).

# Student's $t$ : a story

- W. S. **Gosset**, a chemist at **Guinness**, needed to compare beer batches from **small samples** — and published under the pen-name “*Student*”.
- His  **$t$ -distribution** describes the estimator's behaviour when  $\sigma^2$  is itself estimated and  $n$  is small — exactly our situation.
- *Hold your beer*: it is the workhorse of significance testing.

## The $t$ -statistic: $\hat{\beta}$ in standard-error units

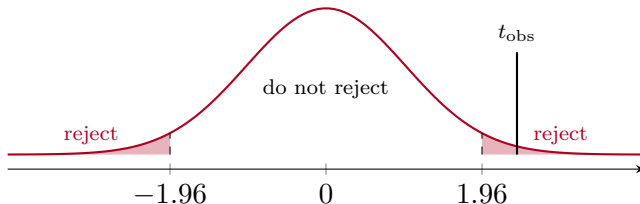
To test  $H_0 : \beta_k = \beta_k^0$  (usually  $\beta_k^0 = 0$ ): how far is  $\hat{\beta}_k$  from the hypothesised value, **measured in standard errors**?

$$t = \frac{\hat{\beta}_k - \beta_k^0}{\widehat{\text{se}}(\hat{\beta}_k)} \xrightarrow{H_0: \beta_k=0} \frac{\hat{\beta}_k}{\widehat{\text{se}}(\hat{\beta}_k)}.$$

- So  $t$  is the estimate in **se units**:  $t = 2$  means  $\hat{\beta}_k$  sits two standard errors from 0. A **large**  $|t| \Rightarrow \hat{\beta}_k$  is far from 0 relative to its noise  $\Rightarrow$  implausible if the true  $\beta_k$  were 0.
- Under the assumptions (normal errors),  $t$  follows **Student's  $t$  with  $n - k - 1$  degrees of freedom**, which  $\rightarrow \mathcal{N}(0, 1)$  as  $n$  grows.
- **Why  $t$ , not the normal?** Because we *estimate*  $\sigma^2$  inside  $\widehat{\text{se}}$ ; that extra uncertainty **fattens the tails** in small samples — Gosset's discovery.



# Reading the $t$ : critical values and rejection



- Under  $H_0$ ,  $t$  is centred at 0. **Reject  $H_0$**  when  $t$  lands in the **tails**, beyond the **critical value**.
- Two-sided 5%: reject if  $|t| > 1.96$  (shaded 2.5% + 2.5%). Other levels: 10%  $\rightarrow$  1.645, 1%  $\rightarrow$  2.58. (*Small  $n$ : the threshold is larger — e.g.  $\approx 2.2$  at 10 d.f.; read it off the  $t$ -table.*)
- **Rule of thumb:**  $|\hat{\beta}_k| > 2 \hat{\text{se}} \Rightarrow$  significant at 5%. (The  $p$ -value, next, is the tail area beyond the observed  $|t|$ .)

## What a $p$ -value is — and is not

The  $p$ -value is the probability, **if  $H_0$  were true**, of a  $|t|$  at least as large as the one observed. A small  $p$  means the data are unlikely under  $H_0$ , so we reject.

It is **not** the probability that  $H_0$  is true.

- **95% confidence interval:**  $\hat{\beta}_k \pm 1.96 \widehat{\text{se}}(\hat{\beta}_k)$ . If it **excludes 0**, reject  $H_0$  at 5% (equivalent to the  $t$ -test).
- **But significance is not effect size**, and insignificance is not “no effect” — see the next slide.

# Significance is not effect size: power and magnitude

## Power — the chance of catching a true effect

The **power** of a test is the probability it rejects  $H_0$  when  $H_0$  is false — i.e. detects a real effect. It **rises** with the sample size  $n$ , the size of the true effect, and precision (small SE).

- **Small  $n \Rightarrow$  low power:** a real, theory-backed effect can look **insignificant** only because the SE is large. *Absence of evidence is not evidence of absence* — get **more data**, don't drop the variable.
- **Large  $n \Rightarrow$  almost everything rejects  $H_0$ :** even economically trivial effects become “significant.”

**Judge the magnitude concretely:** convert  $\hat{\beta}$  to real units (via the functional form), compare it to a **benchmark** (the mean or SD of  $y$ , a policy threshold, other studies), and read the **whole CI** — a wide interval is *uninformative*, not *zero*.

# Reading the whole output: $R^2$ and the $F$ -test

$R^2$  — goodness of fit

Total variation in  $y$  splits into explained and residual parts:

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

the *total* (TSS), *explained* (ESS) and *residual* (RSS) sum of squares.

Then  $R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1]$  — the share of  $y$ 's variance the model explains.

- **High  $R^2 \neq$  good estimate:** a biased  $\hat{\beta}$  can fit well, and a valid causal effect can have low  $R^2$ . *Fit  $\neq$  correctness* (adjusted  $R^2$  penalises extra regressors).
- **The  $F$ -test checks joint significance** ( $H_0$ : several slopes = 0 at once) — also printed in every output. Handy under multicollinearity: individual  $t$ 's look insignificant, yet the joint  $F$  is significant.

## Section 3

The assumptions — and when they fail

# The OLS assumptions (Gauss–Markov)

- 1 **Linearity in parameters:**  $y = X\beta + \varepsilon$ .
- 2 **Exogeneity (zero conditional mean):**  $\mathbb{E}(\varepsilon_i \mid X) = 0$ .
- 3 **Homoskedasticity:**  $\text{Var}(\varepsilon_i \mid X) = \sigma^2$  (constant).
- 4 **No autocorrelation:**  $\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0, i \neq j$ .
- 5 **No perfect multicollinearity** — needed for  $(X'X)^{-1}$  to exist.

(Normality of the errors is added *only* for exact small-sample  $t$ -tests.) Here  $\mathbb{E}[\cdot \mid X]$  is the **expectation** given the regressors; Cov, Var are covariance and variance.

## What each assumption buys

1–2  $\Rightarrow$  OLS is **unbiased & consistent**. Adding 3–4  $\Rightarrow$  it is **BLUE** (minimum variance among linear unbiased estimators). Each "when it fails" slide drops one and asks what breaks.

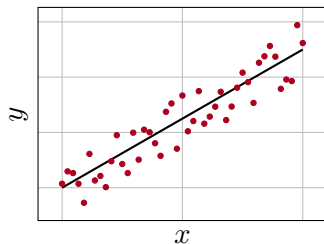
# Heteroskedasticity: a variance problem

Heteroskedasticity is always a problem with the **variance (volatility)** of the forecast errors:

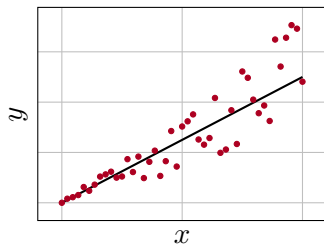
- **Cross-section:** the error variance varies with the explanatory variable  $x_i$ .
- **Time series:** the error variance varies with the explanatory variable  $x_t$ .
- **Spatial (cross-sectional) form:** the error *variance* differs systematically across places — e.g. larger in some regions than others. (*Variance, not correlation — error correlation across units is autocorrelation, next.*)
- **Another time-series form:** the error variance at  $t$  varies with the variance of *past* errors.

# Homoskedasticity vs. heteroskedasticity

Homoskedastic



Heteroskedastic



Left: residual spread is roughly **constant**. Right: it **fans out** with  $x$ .  
Beware — correlated errors can look similar, so rely on **formal tests**, not the eye.



# Heteroskedasticity: detection

$\text{Var}(\varepsilon_i \mid X)$  is **not** constant. Then OLS is still **unbiased** but no longer efficient, and **default standard errors are wrong**  $\Rightarrow$  invalid  $t$ -tests.

## Tests ( $H_0$ : homoskedasticity)

**Breusch–Pagan:** regress  $\hat{\varepsilon}_i^2$  on the regressors,  $\hat{\varepsilon}_i^2 = \delta_0 + \delta_1 x_{1i} + \cdots + v_i$ , and test  $\delta_1 = \cdots = 0$ .

**White:** same idea, adding *squares* and *interactions*.

$p < 0.05 \Rightarrow$  reject homoskedasticity. ( $\hat{\varepsilon}_i = y_i - x_i' \hat{\beta}$  are the **residuals** — the estimated errors.)

# Heteroskedasticity: the fix

## Use robust standard errors by default

Heteroskedasticity-consistent (**HC** / White) standard errors fix inference without changing  $\hat{\beta}$ .

- The **sandwich** covariance (White, 1980):

$$\widehat{\text{Var}}(\hat{\beta}) = (X'X)^{-1} X' \text{diag}(\hat{\varepsilon}_i^2) X (X'X)^{-1},$$

where  $\text{diag}(\hat{\varepsilon}_i^2)$  is the diagonal matrix of squared residuals.

- $\hat{\beta}$  is **unchanged** — only the SE (hence  $t$ ,  $p$ , CIs) are corrected.
- The form shown is **HC0**; finite-sample versions inflate the squared residuals — **HC3** is the recommended default.
- Alternative: **GLS** / **FGLS** model the variance explicitly (less common in practice).

# Autocorrelation: an error-correlation problem

Autocorrelation is no longer about the *variance* of the error, but about the **error itself being correlated** across observations:

- **Cross-section:** one unit's errors affect another's — when units share common characteristics or factors that create dependence.  
*Example: spatial data.*
- **Time series:** the error at  $t$  is a function of past errors  $(t - 1, t - 2, \dots)$ .

# Autocorrelation: tests and fix

Formally,  $\text{Cov}(\varepsilon_i, \varepsilon_j) \neq 0$ :

- **spatial:**  $\varepsilon_i = \rho \sum_{j \neq i} w_{ij} \varepsilon_j + u_i$  ( $w_{ij}$  a proximity weight);
- **time series:**  $\varepsilon_t = \rho_1 \varepsilon_{t-1} + \dots + \rho_p \varepsilon_{t-p} + u_t$ .

OLS stays unbiased, but **standard errors are wrong**.

- **Tests:** Durbin–Watson ( $DW \approx 2$ : none;  $\approx 0$ : positive;  $\approx 4$ : negative — but invalid with a lagged dependent variable); Breusch–Godfrey (higher-order, valid with lagged  $y$ ).
- **Fix: HAC SE** (Newey–West, 1987) — robust to *both* heteroskedasticity and autocorrelation; or Cochrane–Orcutt / Prais–Winsten.

# Multicollinearity

Two or more regressors are strongly linearly related, e.g.  $x_1 \approx a + b x_2$ :

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon.$$

- Consequences: **unstable** coefficients (signs flip), **inflated** standard errors / wide CIs, hard to separate individual effects. ( $\hat{\beta}$  *stays unbiased*.)
- **Detect:** variance inflation factor (VIF). **Fix:** drop or combine variables; get more data.
- *Example:* regressing salary on **age and experience** — highly correlated, so  $\beta_1, \beta_2$  are hard to pin down.

# Non-normality of residuals

- Normal errors are needed for **exact small-sample**  $t$ -tests — **not** for unbiasedness, and not asymptotically (the CLT rescues large  $n$ ).
- **Test:** Jarque–Bera ( $H_0$ : residuals are normal);  $p < 0.05 \Rightarrow$  reject.
- If violated **and**  $n$  small: use the **bootstrap** — resample the data with replacement many times to approximate the sampling distribution of  $\hat{\beta}$ .

# Endogeneity: the deepest problem

When  $\mathbb{E}(\varepsilon \mid X) \neq 0$ , OLS is **biased** — and no robust SE can fix that.  
Three sources:

- **omitted variables, reverse causality, measurement error.**

**Which way does omitted-variable bias go?** Its sign is (effect of the omitted variable on  $y$ )  $\times$  (its correlation with  $x$ ). *Wage gap*: education/ability raise wages and correlate with the woman dummy in the raw data, biasing the uncontrolled gap — adding controls moved  $-0.285 \rightarrow -0.10$ .

## Causal inference (the rest of the course)

Designed to defeat the first two. Four workhorses: randomized experiments, **difference-in-differences**, regression discontinuity, and **instrumental variables** (later: shift-share).

## Section 4

# Estimating in Python



# The toolkit: which package, for what

| package     | use it for                                                                           |
|-------------|--------------------------------------------------------------------------------------|
| pandas      | load, inspect, clean, merge, transform data (the <code>DataFrame</code> )            |
| numpy       | numerical operations (logs, powers, arrays)                                          |
| statsmodels | <b>estimation + inference</b> (coef, se, $t$ , $p$ , CI) and <b>diagnostic tests</b> |
| matplotlib  | plots and visual checks                                                              |

Typical workflow: **load**  $\rightarrow$  **inspect**  $\rightarrow$  **transform**  $\rightarrow$  **estimate**  $\rightarrow$  **robust SE**  $\rightarrow$  **diagnostics**. (Fixed effects and IV come later, with `pyfixest`.)

# Load and inspect (pandas)

```
import pandas as pd

df = pd.read_stata("wage.dta")          # or read_csv / read_excel
df.head(); df.describe(); df.info()    # inspect first!
df = pd.merge(df, dr, on="id", how="inner")
```

- **Always inspect before estimating:** types, ranges, missing values, oddities.
- **merge:** `inner` keeps matched rows, `outer` keeps all (gaps  $\rightarrow$  NaN), `left/right` keep one side. Windows paths: prefix with `r"..."`.

# Transform variables (pandas + numpy)

```
import numpy as np

df["lwage"] = np.log(df["wage"])      # logs via numpy
df["exper2"] = df["exper"]**2        # quadratic term
```

- For **dummies/categoricals**, don't recode by hand — let the formula do it with `C(...)`: e.g. `C(sector)` expands a category into dummies (one level kept as baseline).

## Estimate and read the output (statsmodels)

```
import statsmodels.formula.api as smf

f = "lwage ~ woman + educ + exper"
m = smf.ols(f, df).fit()
print(m.summary())      # coef, se, t, p-value, 95% CI
```

- Formula syntax:  $\sim$  splits  $y$  from the regressors;  $x_1*x_2$  adds both terms **and** their interaction;  $C(g)$  makes dummies.
- Each summary row is exactly what we covered:  $\hat{\beta}$ , its se, the  $t$ -stat, the  $p$ -value, and the 95% CI.

# Best practice: robust standard errors by default

```
# reuse the formula f from the previous slide
m = smf.ols(f, df).fit(cov_type="HC3")    # robust SE
m = smf.ols("y ~ x", df).fit(cov_type="HAC",
                             cov_kwds={"maxlags": 4})    # time series
```

- `cov_type="HC3"` leaves  $\hat{\beta}$  unchanged and corrects se,  $t$ ,  $p$ , CI — make it your **default** in cross-sections.
- For time series use **HAC** (Newey–West); set `maxlags` from the data frequency.

## Diagnostics, when you need them (statsmodels.stats)

```
from statsmodels.stats import diagnostic as smd
from statsmodels.stats.stattools import jarque_bera

_, p_white, _, _ = smd.het_white(m.resid, m.model.exog)
_, p_bp, _, _ = smd.het_breuschpagan(m.resid, m.model.exog)
jb, p_jb, *_ = jarque_bera(m.resid) # normality
```

- With robust SE already the default, you test mainly to **understand** the data — and to decide whether time-series data need **HAC**. A small  $p$  flags the problem.
- **Don't test-then-switch:** pre-testing distorts inference; defaulting to robust SE avoids it. (`het_white` can fail with many regressors — fall back to Breusch–Pagan.)

The single text for this course — read it for the full, detailed bibliography behind everything in these slides:

Wooldridge, Jeffrey M. 2025. *Introductory Econometrics: A Modern Approach*. 8th ed. Cengage Learning.